

DECISION-SUPPORT SYSTEM • INTEGRATING DATA AND AI FOR A SUSTAINABLE PAHANG

NADA DASHBOARD

NEW-ERA ASSISTANCE DISTRIBUTION AGILITY

PAGE 1

Employment & Poverty

Explore employment, income and poverty patterns across Pahang's districts and identify areas where low unemployment may still hide high poverty.

PAGE 2

District Vulnerability

Compare districts by their overall vulnerability and understand the key factors contributing to each district's risk.

PAGE 3

Economic Outlook

View estimated changes in household income and poverty, including the level of uncertainty around each estimate, to support earlier planning and monitoring.

Infographic

“When Employment Looks Healthy, Who Is Still Struggling?”

Monitor district-level economic vulnerability, labor market decoupling, and inter-survey income shifts across Pahang's 11 districts to support evidence-based resource allocation, targeted aid distribution, and localized infrastructure prioritization between official survey releases

LOW UNEMPLOYMENT DOES NOT ALWAYS MEAN LOW ECONOMIC HARDSHIP.

LOW UNEMPLOYMENT
= NOT NECESSARILY LOW POVERTY



VS

MULTIDIMENSIONAL VULNERABILITY
= A BETTER SIGNAL OF REAL HARDSHIP



ABOUT THIS PROJECT



NADA uses data and AI to uncover hidden vulnerabilities in Pahang by combining socioeconomic indicators into a single, clear signal – the District Vulnerability Index (DVI).

KEY NUMBERS

11	Administrative Districts	2019, 2022, 2024	HIES Benchmark Years
55	District-Year Observations	Objective	Identify vulnerable districts between HIES releases
33	Matched HIES District-Years		
2020-2024	DVI Panel		

1 THE CHALLENGE



“Poverty and income are not measured frequently enough, but decisions still need to be made every year.”

This project addresses the gap between infrequent HIES releases and more frequent data (labour, amenities, demographic, education and inflation to cross-sectionally) to create a timely district-level vulnerability signal.

2 THE KEY DISCOVERY

A. UNEMPLOYMENT DECOUPLES FROM POVERTY

UNEMPLOYMENT ↔ POVERTY

Pearson r	=	-0.020
Spearman p	=	-0.038
p-value	>	0.8
OLS R ²	=	0.000

Unemployment is not a reliable standalone measure of poverty.

B. LFPR IS MORE INFORMATIVE

LFPR ↔ POVERTY

Pearson r	=	-0.564
p-value	=	0.0006
R ²	=	0.318

Labour Force Participation Rate (LFPR) provides a stronger signal of hardship.

WORKING-POOR SPOTLIGHT

LIPIS
Low unemployment, high poverty

MARAN
Low unemployment, high poverty

Lipis and Maran illustrate the working-poor problem



3 NADA ANALYTICS ENGINE

OpenDOSM / MOE DATA

Labour, poverty, amenities, educational, inflation.

EMPLOYMENT-ARKEYITS

Test whether contemporary reflects hardship.

DVI (dimension-based)

Multidimensional transparency.

NOWCASTING ENGINE

Predict present-day poverty & income using AI models.

INTERACTIVE DASHBOARD

Visualize, explore and monitor district vulnerability.

SMARTER AID & PLANNING DECISIONS

Prioritize, allocate and evaluate impact better.

A. EMPLOYMENT-POVERTY ANALYSIS

Systematically tests whether unemployment reflects household hardship.

B. DVI - DISTRICT VULNERABILITY INDEX

IPCC-inspired framework

LABOUR-MARKET, HOUSEHOLD ECONOMIC PROXY, LIVING-CONDITION, BRIDGE, RM, PRICE PRESSURE

C. PRIORITY NOWCAST WATCH-LIST

Support signal

1	MARAN	DVI 0.758
2	LIPIS	DVI 0.743
3	JERANTUT	DVI 0.704

OTHER PRIORITY DISTRICTS

ROMPIN, LIPIS, BERA, PEKAN

These districts should be prioritized for monitoring, as they are in the red zone of the DVI.

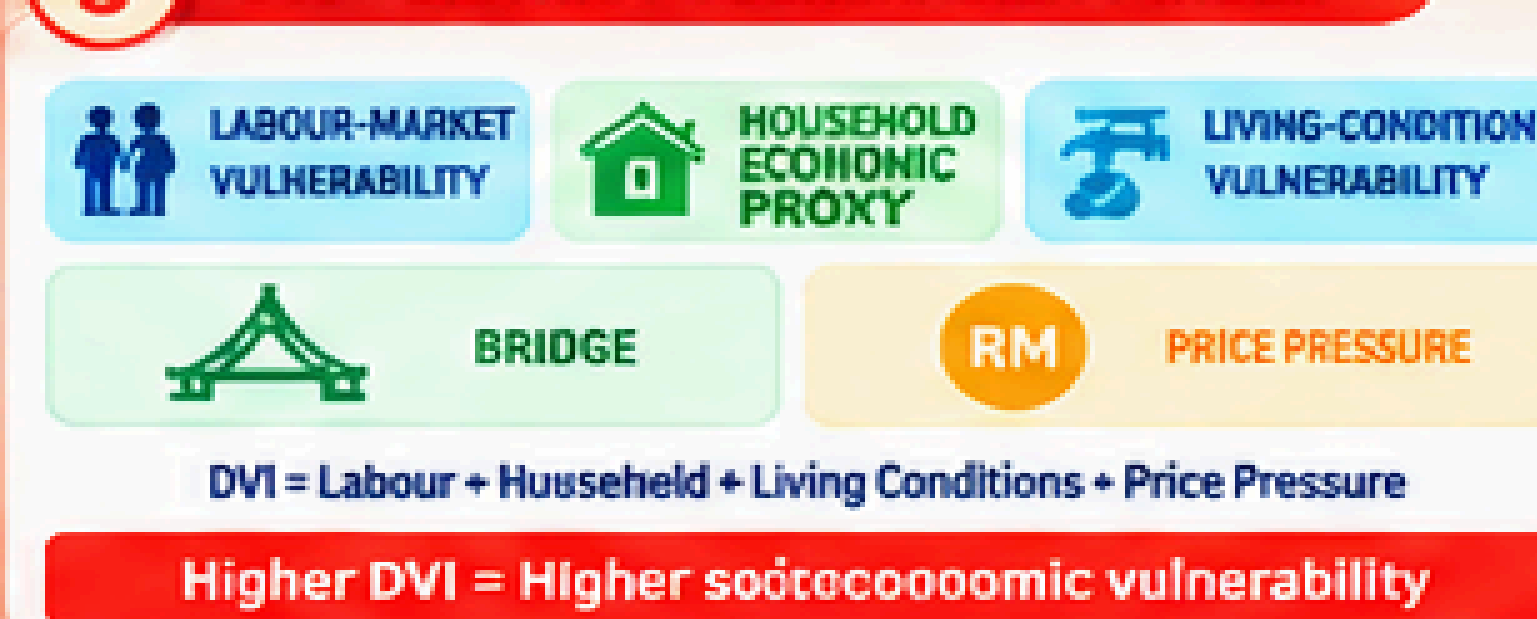
4 EMPLOYMENT-POVERTY ANALYSIS

Systematically tests whether unemployment reflects household hardship.

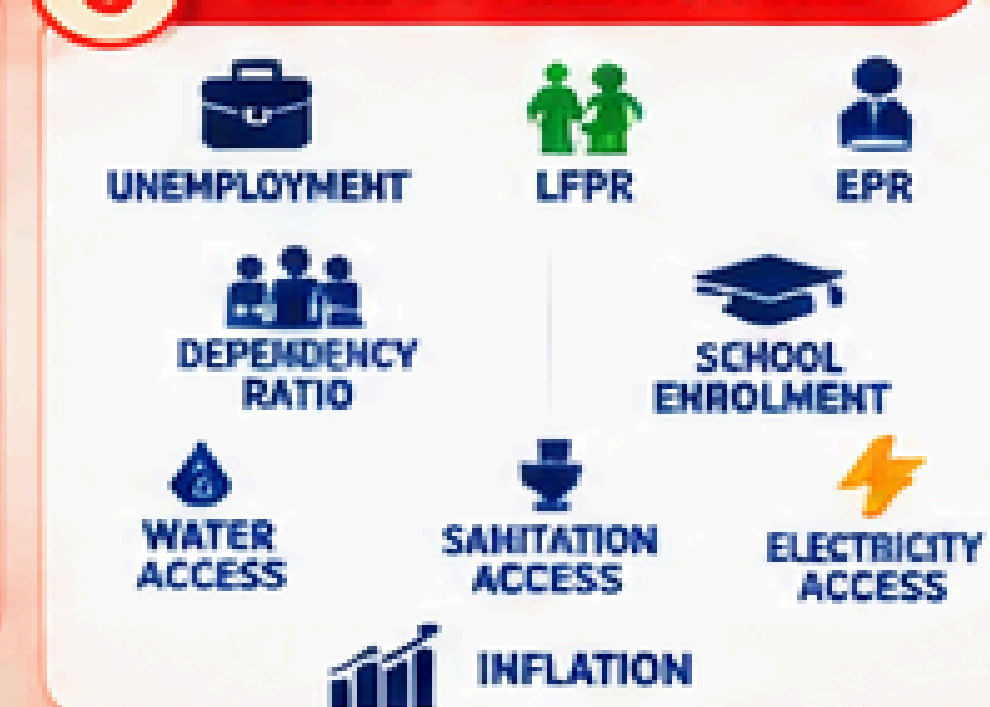
UNEMPLOYMENT → X → POVERTY

Evidence shows unemployment is not a reliable indicator of poverty in Pahang's districts

5 DVI - DISTRICT VULNERABILITY INDEX



6 CORE DVI INDICATORS



7 REGRESSION TREE FOR INCOME MODEL

MODEL PERFORMANCE

R² = 0.662
MAPE = 6.31%

TOP FACTORS (Example)

- LFPR
- Education Level
- Sewerage Access
- Electricity Access
- Dependency Ratio

Income nowcasting is more reliable than poverty nowcasting.
All forecasts are decision-support estimates, not official statistics.

8 POTENTIAL IMPACT FOR PAHANG

SOCIAL
Better targeting of vulnerable households, including overlooked working-poor communities.

ECONOMIC
More efficient allocation of state funds, welfare assistance and district-level development resources.

ENVIRONMENTAL / INFRASTRUCTURE
Support targeted investment in basic services and local infrastructure where deprivation is a key vulnerability driver.

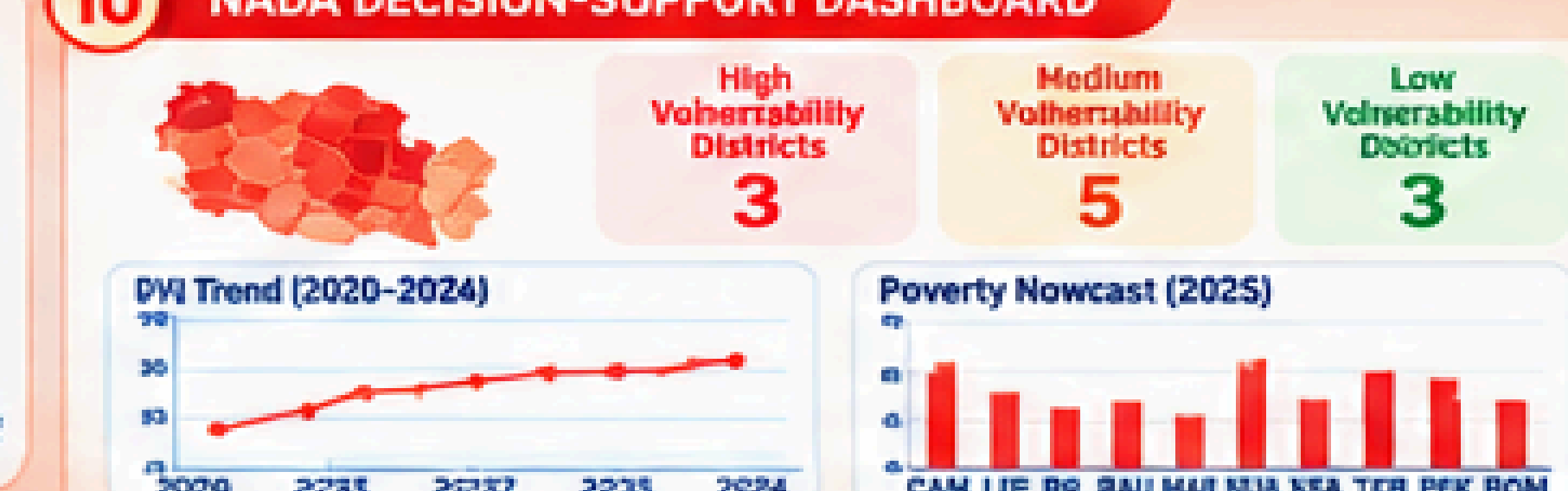
12 ORGANISED BY



9 FROM DATA TO DECISION

Which district is struggling?
Is unemployment telling the full story?
What is the district's DVI level?
What are the main vulnerability drivers?
What do the nowcasted poverty and income estimates suggest?
Which districts should receive priority support?

10 NADA DECISION-SUPPORT DASHBOARD



Executive Overview & Quick Insight | District Vulnerability Index | Economic Nowcasting & What-if | Scenario Simulation

FILTERS (Example): District | Year | Vulnerability Level

SUPPORTING THE SDG



GO BEYOND UNEMPLOYMENT.
MEASURE VULNERABILITY BETTER.

FORECAST EARLIER. ALLOCATE SMARTER.

NADA transforms fragmented socioeconomic data into a transparent, updatable decision-support system for targeting aid, strengthening resilience and improving aid distribution in Pahang.





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DISTRICT VULNERABILITY
INTELLIGENCE — PAHANG



Overview & Labour
Mismatch

District Vulnerability
Index (DVI)

Nowcasting Engine

Definition

How to read

Youth Action
Guide

Data Sources: Department Of
Statistics Malaysia (DOSM)

Executive Overview & Labour Mismatch

Is low unemployment hiding poverty — and what does this mean for Pahang's young population? This page surfaces the gap between headline labour statistics and household-level economic health across Pahang's 11 districts

HIGHEST DISTRICT RISK

Maran

Score: 0.99

STATE AVERAGE
POVERTY RATE

6.22%

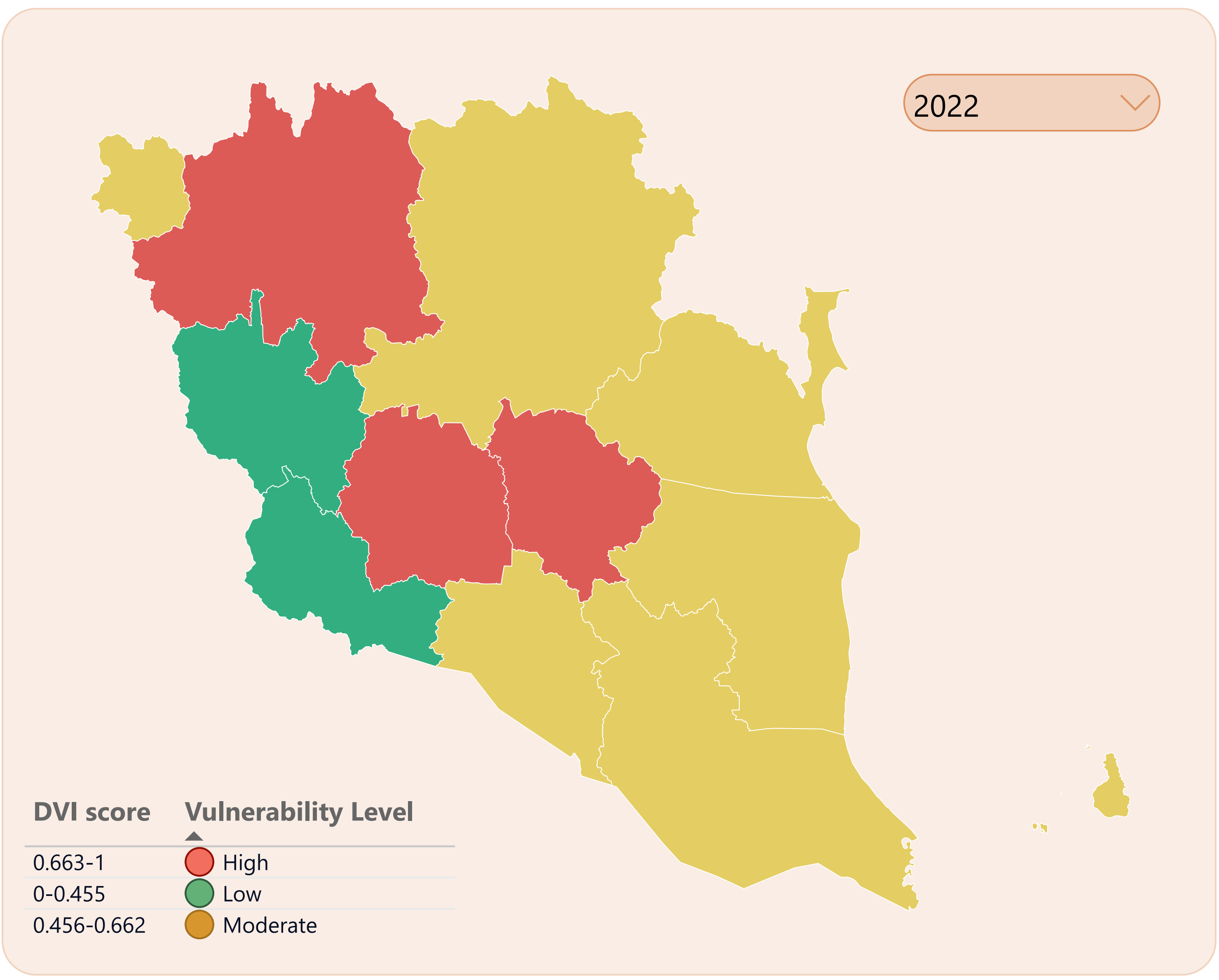
STATE AVERAGE LFPR

63.85%

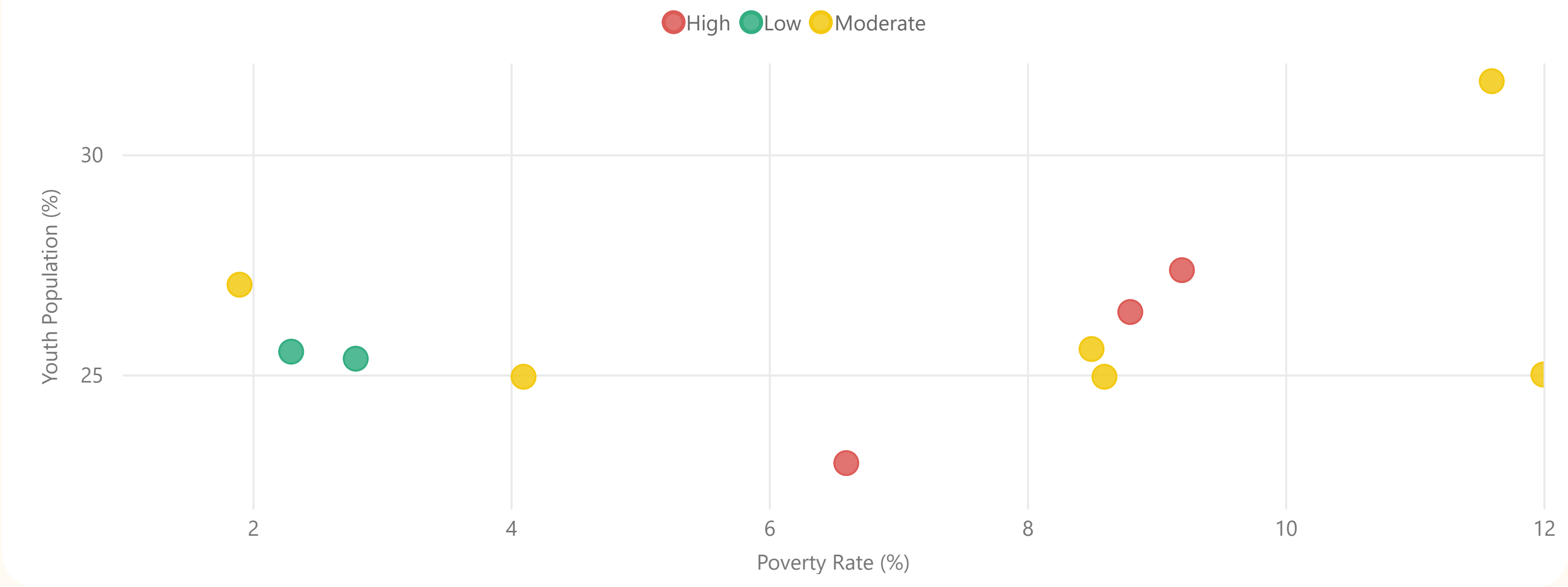
STATE AVERAGE YOUTH
POPULATION

25.78%

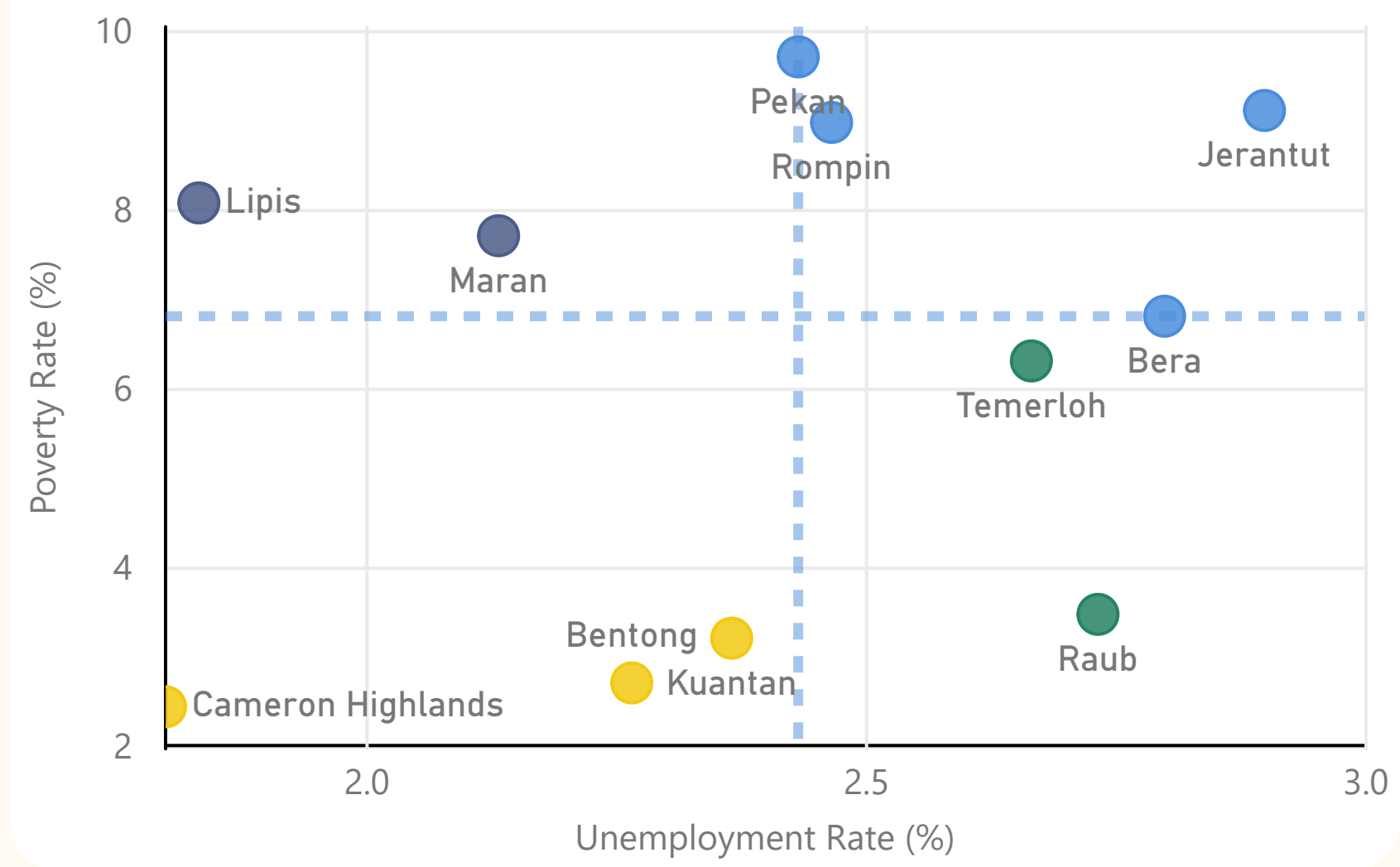
Pahang District Vulnerability Map



Youth Population vs. Poverty



Unemployment vs. Poverty



Unemployment vs. Poverty

- Low unemployment does not always mean low poverty.
- Lipis and Maran have low unemployment but higher poverty.

Youth Population vs. Poverty

- Districts with more young people do not always have higher poverty.
- Youth population alone cannot explain poverty.

District Vulnerability Index (DVI)

HIGHEST DISTRICT RISK

Maran

Rank #1

DVI SCORE

0.93

Higher = Greater vulnerability

VULNERABILITY LEVEL

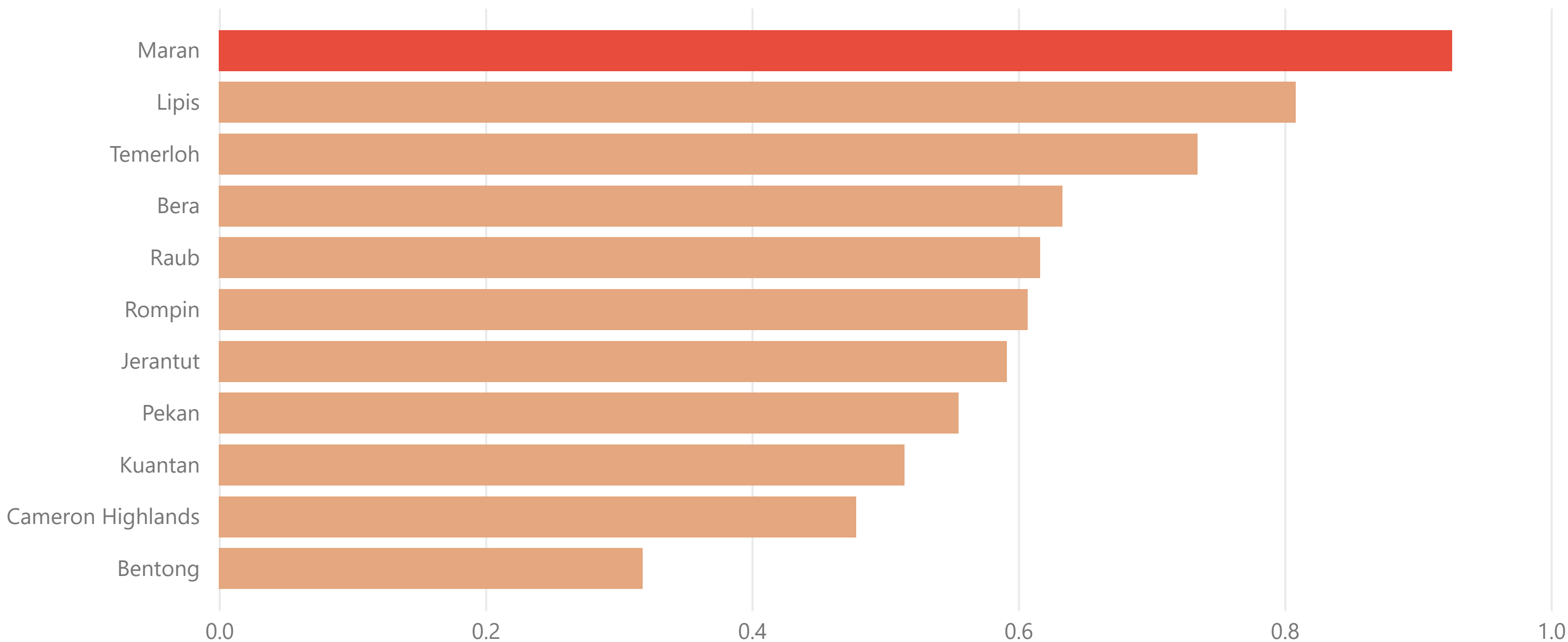
High

MAIN DRIVER

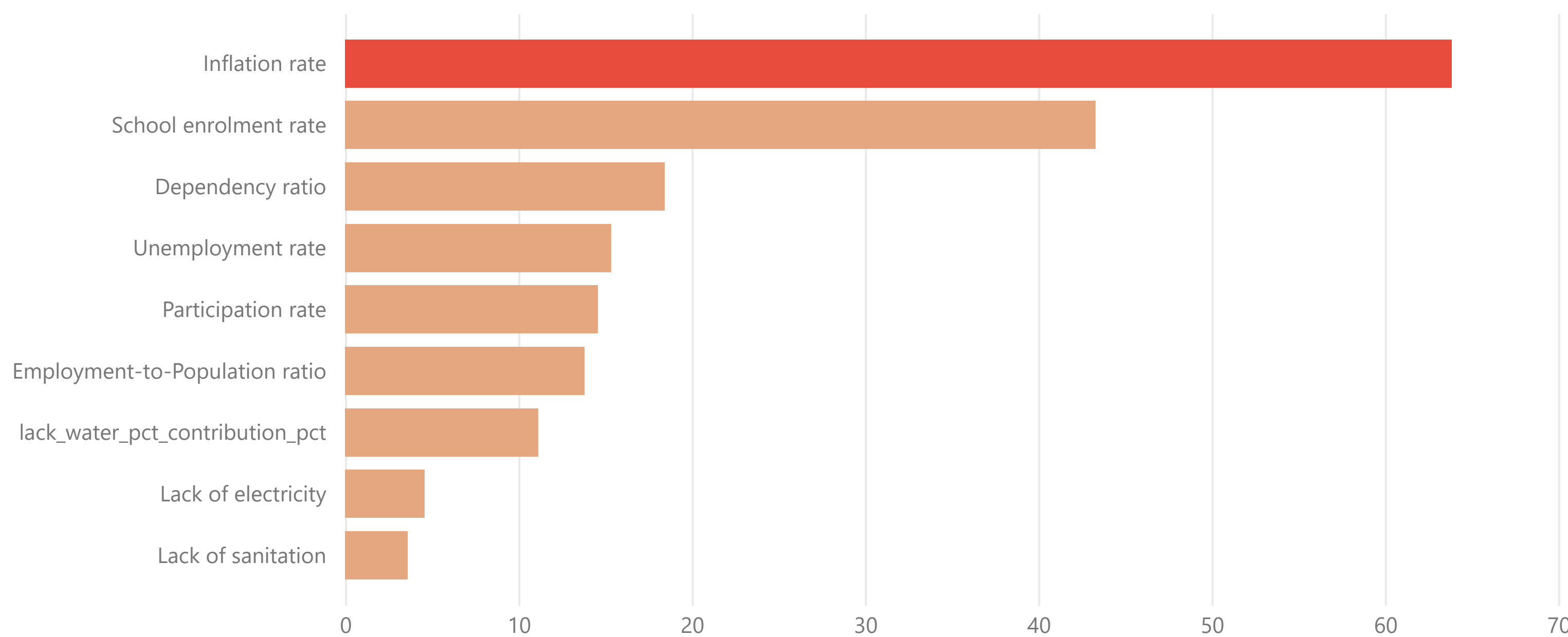
Inflation

Largest contributor to DVI

DVI Ranking Leaderboard



Indicator-DVI Correlation



Policy Action Summary

Year : 2023

District : All

District	Year	Vulnerability Level	Dominant Pillar	Dominant Indicator	Dominant Contribution
Lipis	2023	High	Exposure	Inflation	38.19
Maran	2023	High	Exposure	Inflation	100.00
Temerloh	2023	High	Exposure	Inflation	63.00
Bentong	2023	Low	Lack of Adaptive Capacity	School Enrolment	34.37
Raub	2023	Low	Lack of Adaptive Capacity	School Enrolment	75.47
Bera	2023	Moderate	Exposure	Inflation	48.73
Cameron Highlands	2023	Moderate	Exposure	Inflation	64.51
Jerantut	2023	Moderate	Lack of Adaptive Capacity	School Enrolment	49.63
Kuantan	2023	Moderate	Exposure	Inflation	89.95



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How to read

Explore why

Data Sources: Department Of
Statistics Malaysia (DOSM)

Vulnerability Typology

THIS DISTRICT'S MAIN ISSUE

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TYPES OF STRUGGLE FOUND

3

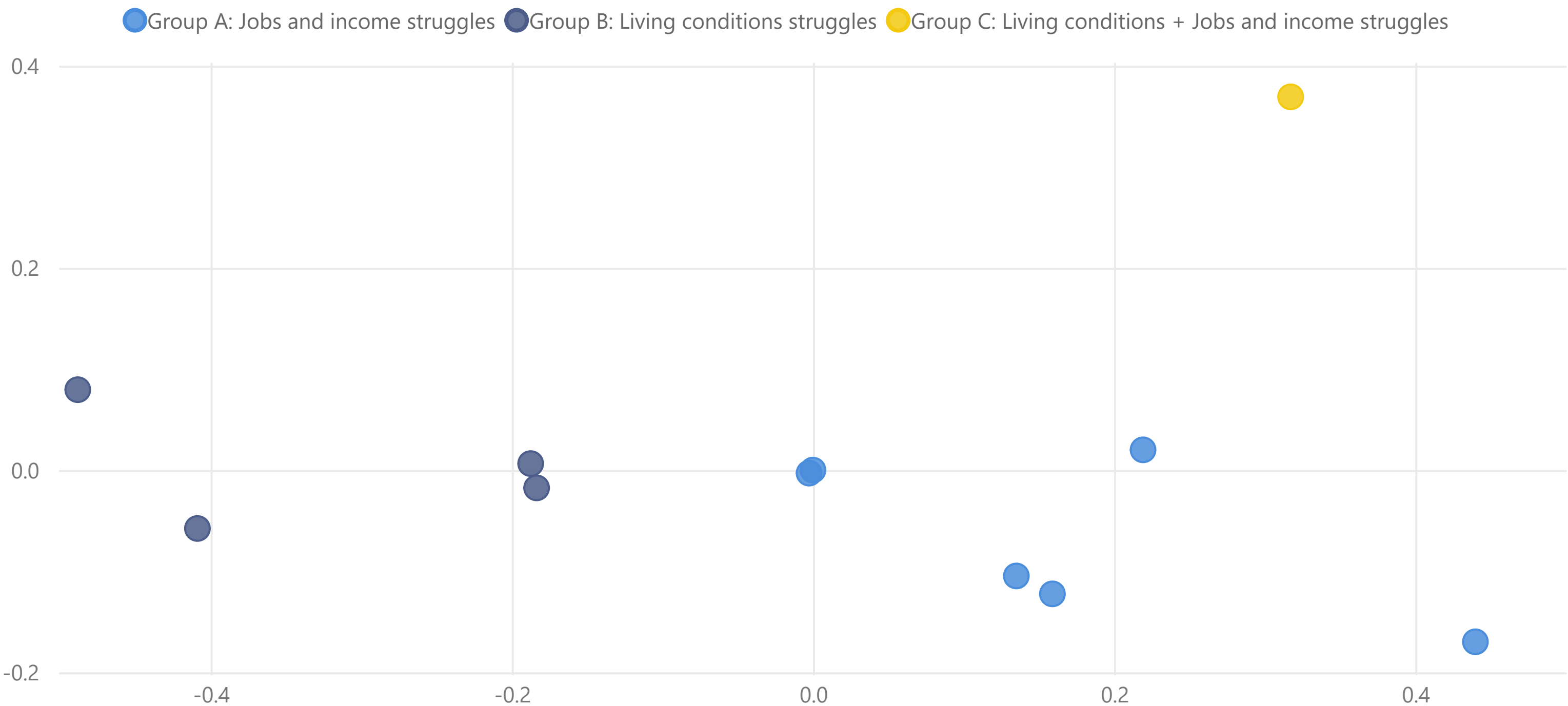
CHECKED TWO DIFFERENT WAYS

100%

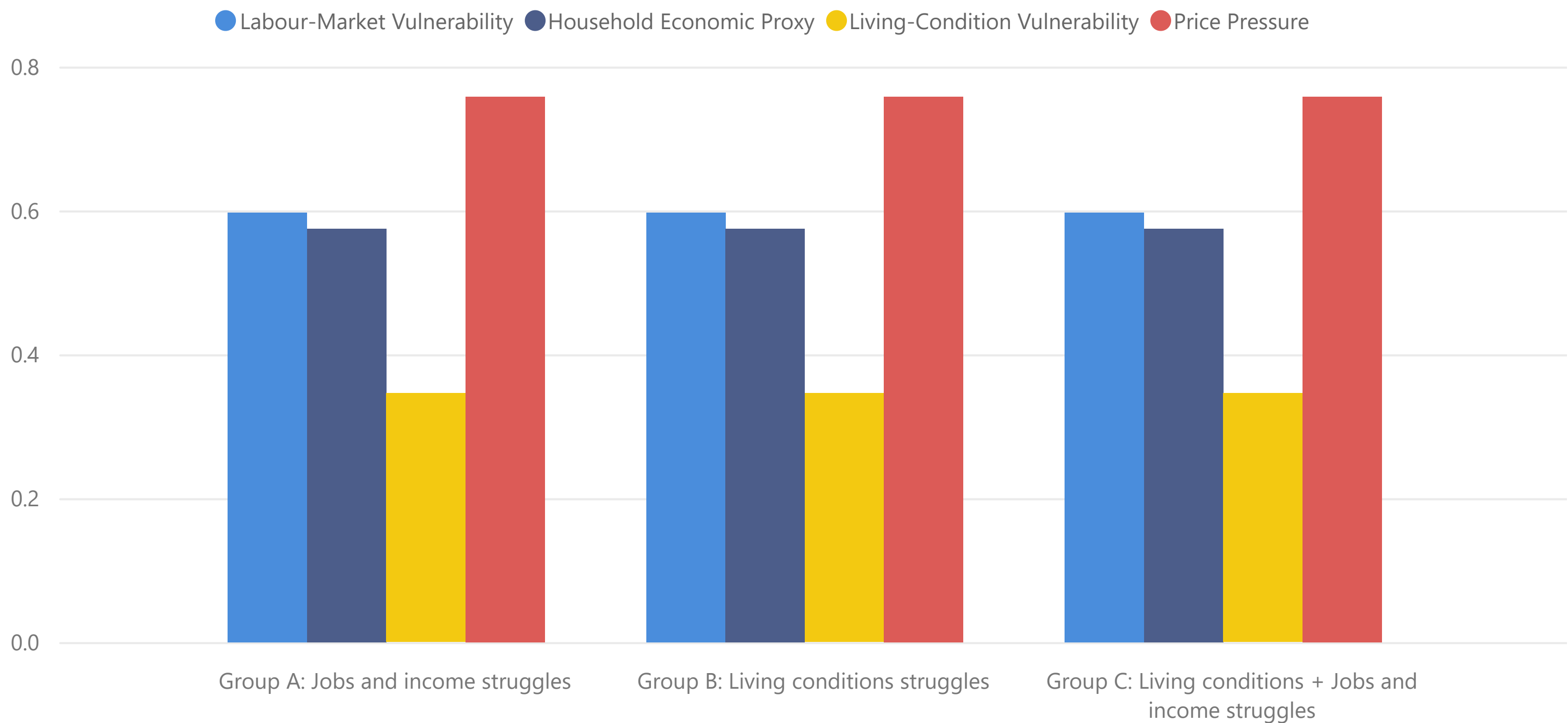
DISTRICTS NEEDING MORE DATA

4

District Vulnerability Typology



Dominant Vulnerability Drivers by Cluster



Policy Action Summary

All

District	Group	Main Issue	Data Available
Bentong	Group B: Living conditions struggles	No single issue stands out	2
Bera	Group A: Jobs and income struggles	Mainly a jobs and income issue	3
Cameron Highlands	Group B: Living conditions struggles	Mainly a living conditions issue	3
Jerantut	Group A: Jobs and income struggles	Mainly a jobs and income issue	2
Kuantan	Group B: Living conditions struggles	No single issue stands out	2
Lipis	Group C: Living conditions + Jobs and income struggles	Mainly a living conditions issue	3
Maran	Group A: Jobs and income struggles	Not enough data yet — treat with caution	1
Pekan	Group B: Living conditions struggles	No single issue stands out	3
Raub	Group A: Jobs and income struggles	Mainly a household finances issue	3



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Economic Nowcasting & Uncertainty Simulation

Estimated present-day district median income and poverty rate for non-survey years, with clear communicated confidence ranges

DISTRICT :

All

MODEL RELIABILITY :



High

Income



Moderate

Poverty

Pahang Est. Median Income (2026)

RM4,600.70

CI: RM4040.76 - RM4933.45

Pahang Est. Poverty Rate (2026)

7.62%

CI: 6.18% - 10.21%

Model Reliability

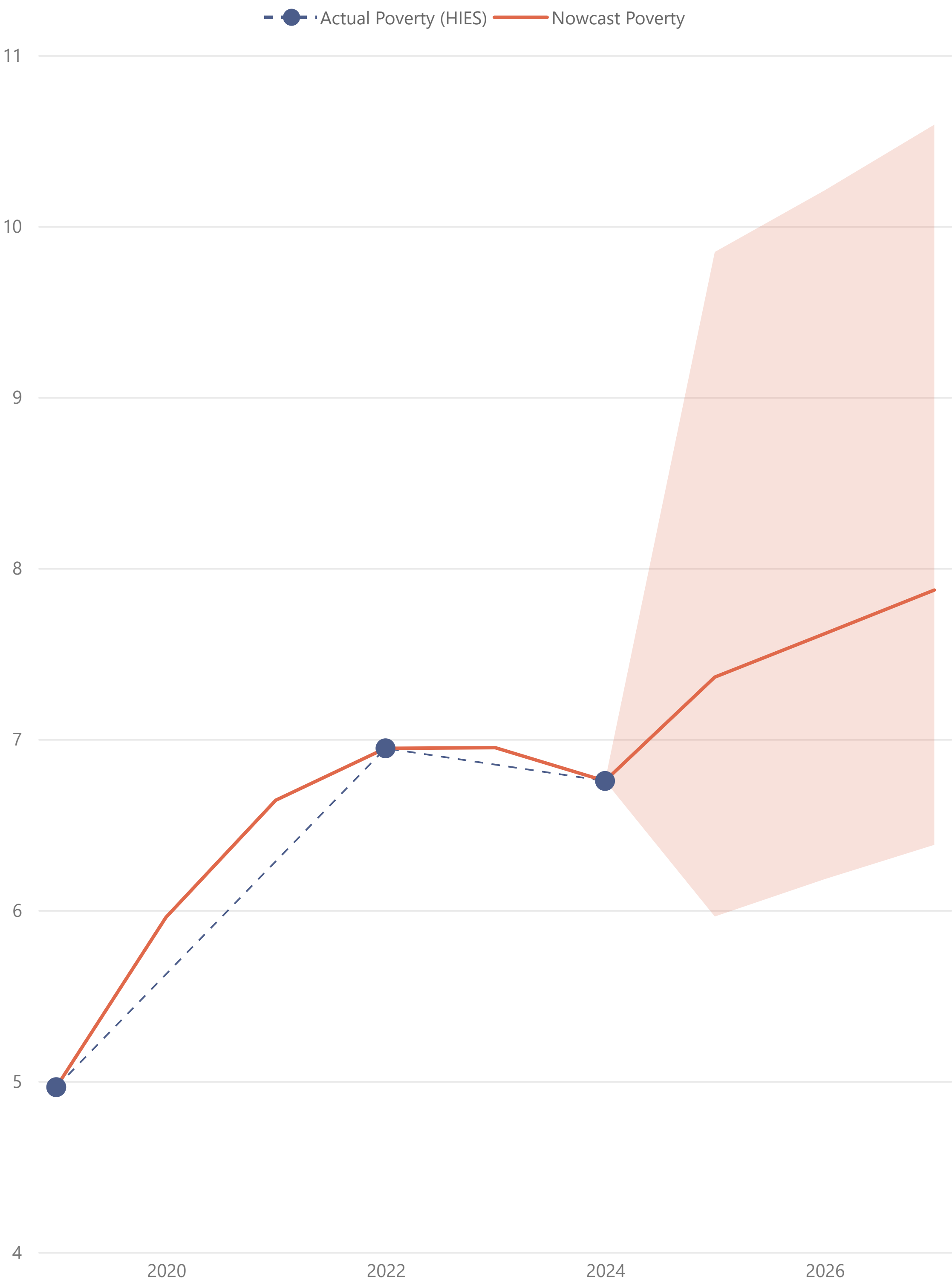
Moderate-to-High

Income: stronger predictive performance

Poverty: for identifying general trends

Metric	Income	Poverty
LOOCV MAE	278.45	1.80
LOOCV RMSE	356.26	2.17
Validation MAPE	6.31	48.07
Validation R ²	0.66	0.53

Nowcasted Poverty Rate



Nowcasted Median Income

